



CIS 4951

Capstone II

Section: RVC

Internet/Fully Online

Fall Term 2026

Professor Information

Masoud Sadjadi

Roles: Primary Instructor

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Office Hours: Monday to Friday, 1:00–2:00 PM Eastern, on Zoom — drop in,

[Private Zoom meeting access removed from public copy]

appointment,

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Website: <https://www.cis.fiu.edu/~sadjadi/>

Department or Academic Unit: Knight Foundation School of Computing & Information Sciences

Additional Notes

Please do not email me about this course. Course email gets lost among several hundred other messages a week, and you will get a slower answer than you deserve. Use one of these instead:

If you want...	Go to
A question for your team	Your team channel on the capstone site
A question for your Product Owner	Your team channel — your team leader raises it, not fifteen separate messages
A question for me about the course, an assignment, a grade, or a problem	Canvas Inbox — Inbox -> Compose -> this course -> my name
Something urgent — and only if it is urgent	[Mobile number removed] and say who you are

Canvas is the right place for almost everything. It keeps your question attached to this course, it does not get buried, and I can answer it properly rather than in a hurry between other things.

The mobile number is real and I mean it — for urgent things only. Urgent means you are ill, something has happened at home, or the deadline is tonight and you are stuck.

Course Description

Capstone 2 is the second half of the senior capstone sequence and the term you build. You are placed on a real project — most of them proposed by a company, a lab, or a faculty member outside this course — with a team of three to five students from your own degree program, and you stay with that team and that project for the whole semester.

The work runs as five two-week sprints, and each one has a purpose. The first asks whether the problem is real. The second asks what you are

building and why that way. The third builds the core of it. The fourth measures it against a criterion you stated in advance. The fifth hardens it, hands it over, and states honestly what it does not do.

The term ends in public. Every team presents at the Capstone Showcase on **Friday 4 December**, where industry judges score the work live. One week later, you become the judges yourself, at the Capstone 1 Virtual Showcase on **Friday 11 December**.

This course is delivered entirely online. There are no in-person class sessions.

Your registration may show a meeting time and a room. **No class is held there.** All coursework is completed remotely through the FIU Capstone site. Team stand-ups run twice a week on times your team sets from its own availability.

One exception: the Capstone Showcase on Friday 4 December is in person, and every team presents.

(U01:) Students in this section are **required to attend in person.**

(RVC:) Students in this section are **strongly encouraged** to attend — it is the best hour you will spend on this course — but in-person attendance is **not required** for this section.

The Capstone 1 Virtual Showcase on **Friday 11 December**, where you are a judge, is **online for everyone.**

Teams may meet whenever they wish. **Decisions reached in a meeting must be posted back to the team channel**, so that teammates who could not attend are not excluded.

Student Learning Outcomes/Objectives

- **Analyse a complex computing problem and apply principles of**

computing to identify solutions.

- Design, implement and evaluate a computing-based solution to meet stated requirements in the context of the program's discipline.
- Communicate effectively in a variety of professional contexts
- Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
- Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
- [CS] Apply computer science theory and software development fundamentals to produce computing-based solutions. [CY] Apply security principles and practices to maintain operations in the presence of risks and threats. [DS] Apply theory, techniques, and tools throughout the data science lifecycle and employ the resulting knowledge to satisfy stakeholders' needs. [IT] Use systemic approaches to select, develop, apply, integrate, and administer secure computing technologies to accomplish user goals.

Course Prerequisites

Course prerequisites, if any, are listed below.

Prerequisite: CIS 3950 and Senior Standing.

Expectations of this Course

Two credit hours. Expect 5 to 6 hours per week, including stand-ups, board

upkeep, the build itself, and the sprint review. That is roughly double Capstone 1, which is what two credits means. If a week takes materially longer than that, tell me — the assignment is wrong, not you.

Late work

The course week runs Monday to Friday, and everything is due Friday at 11:59 PM Eastern.

If your week went wrong, take the weekend. Anything posted by Sunday at 11:59 PM Eastern counts in full, with no penalty. You do not have to ask and you do not have to explain.

Work is graded on Monday mornings. Four dates have no weekend because they are events or fixed publications rather than deadlines: **Showcase Preparation, Wednesday 2 December** (Capstone 1 students evaluate from your materials and need them published), the **Capstone Showcase, Friday 4 December**, the **Capstone 1 Virtual Showcase, Friday 11 December**, and the **Final Deliverable, Monday 14 December**. Your three evaluations from 11 December are themselves due Sunday 13 December, with the normal grace.

If you cannot finish even with the weekend, something real has happened and I want to know. **Send me a message in Canvas** — Inbox -> Compose -> this course -> my name. **Please do not email me about this course**; course email is much slower to reach an answer.

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Attendance

There are no class sessions to attend.

Your team's twice-weekly stand-ups are scheduled on your team's own availability, and taking part in them is part of your participation score in every sprint. They are not attendance in the registrar's sense; they are how the sprint is worked.

The one date this course asks you to be somewhere is the **Capstone Showcase, Friday 4 December**. Every team presents. (*U01:*) Attendance in person is required. (*RVC:*) Attendance in person is strongly encouraged but not required for this section.

Use of artificial intelligence

You are expected to use AI tools in this course. Using them is not cheating. Concealing that you used them is.

Every submission carries a four-line record: which tools you used and for what, what you kept, what you checked, and **what the tool got wrong**. The last line is the one that matters — a record that never finds an error is evidence you are not

checking.

In week 13 your team files a **Provenance & Ethics Clearance** covering the whole project: what came from where, what licence it carries, what data was used, and what was checked. It is worth 40 points and it ships inside your final package.

You remain responsible for everything you submit. "The model said so" is not a defence for a fabricated citation, an unlicensed library, or a claim you cannot explain.

Academic integrity

All work is governed by the FIU Student Conduct and Honor Code. Beyond it, two rules specific to this course:

Public or synthetic data only, unless your sponsor's own agreement says otherwise in writing — and where one does, it is explained to your team by name in week two. Nothing live, nothing proprietary, nothing identifying a real person, including material from your own workplace.

You are never required to sign anything to take this course. Where a sponsor asks for a non-disclosure agreement, it is voluntary, you sign it directly with the company and not with the University, and a non-proprietary alternative is always available. **You keep your work and you may publish it.**

Course interaction runs through the capstone site. You will not be given classmates' email addresses or ID numbers, and you should not ask for them.

Accessibility

Students with disabilities should contact the Disability Resource Center. Any accommodation letter will be honoured in full. If any element of this course is difficult to access for any reason, letter or not, **send me a message in Canvas** and we will fix it.

Textbook and Course Materials

Essential Scrum: A Practical Guide to the Most Popular Agile Process - 2012, 1st Edition, Paperback

Required/Recommended: Recommended

Authors: Kenneth S. Rubin

Publisher: Addison-Wesley

Publication Date: 2012

Copyright Date: 2012

ISBN 10: 9780137043293

ISBN 13: 9798375994147

Chapters/Pages: All

Panther Book Pack

Get all required course materials for \$22 per undergrad credit hour through Panther Book Pack. You'll be charged automatically unless you opt out within 3 days after the add/drop deadline.

For more details, to compare costs, and to learn how to access your course materials, visit the [Panther Book Pack information page on FIU OneStop](#).

Assignments & Assessments

Everything in this course is submitted on the capstone site — capstone.cs.fiu.edu — not in Canvas. Canvas holds your assignments and your grades. The work itself lives on the capstone site: your team channel, your sprint posts, your project. Nothing in this course is uploaded to Canvas, which is why every Canvas assignment is set to "No Submission."

The course is worth 1000 points. There is no curve and no extra credit.
Every point available is in the table below.

The fifteen graded items

#	Assignment	Points	Due
1	Week 1 · Course Orientation Wizard	30	Fri 28 Aug
2	Week 1 · Intake Wizard	20	Fri 28 Aug

#	Assignment	Points	Due
3	Week 1 · Project Review & Bid Wizard	60	Fri 28 Aug
4	Week 1 · Team Formation Wizard	40	Fri 28 Aug
5	Week 2 · Cast the Problem & Semester Plan	60	Fri 4 Sep
6	Sprint 1	50	Fri 18 Sep
7	Sprint 2	75	Fri 2 Oct
8	Sprint 3	100	Fri 16 Oct
9	Sprint 4	125	Fri 30 Oct
10	Sprint 5	150	Fri 13 Nov
11	Provenance & Ethics Clearance	40	Fri 20 Nov
12	Showcase Preparation	20	Wed 2 Dec
13	Capstone Showcase — present	100	Fri 4 Dec
14	Capstone 1 Virtual Showcase — judge three projects	30	Fri 11 Dec
15	Final Deliverables — team leader submits	100	Mon 14 Dec
	Total	1000	

Week one is 150 points, and it cannot be made up

The four wizards in week one are worth more than any single sprint except the last one, and they are the only assignments in this course with no second chance. Teams and projects are announced on Monday 31 August and they are built from what you submit that week. A student who does not complete them is placed last, from their intake alone, onto whatever is left.

The five sprints, and why they are priced differently

The sprints are two weeks each, with the same team on the same project, and they ramp: 50 · 75 · 100 · 125 · 150. Sprint 5 is worth three times Sprint 1. All five count and none is dropped.

The ramp is deliberate and it is the forgiveness built into this course. Sprint 1 is the sprint you are least equipped to do well, so it costs the least. By Sprint 5 you have run the ceremonies four times, and that sprint carries the hardest outcome evidence your degree asks for.

Each sprint also has a distinct purpose, and your sprint page tells you which one you are in:

- 1. Is the problem real, and can we run? — establish the cadence, test the problem against your Product Owner and the real data, and build a thin slice that runs end to end.**
- 2. What are we building, and why this way? — choose the approach, and record the alternatives you rejected and the criterion you used.**
- 3. Does the core thing work? — build the central capability on real public or synthetic inputs.**
- 4. Is it any good, measured against a criterion stated in advance? — this is the sprint where *evaluate* is the verb.**
- 5. Can someone else run it, and are we honest about what does not**

work? — harden it, hand it over, and state its limitations before a judge finds them.

How team work is scored: 60 / 40

Every sprint, and the week-two casting assignment, is scored in two parts — 60% the team's artifact and 40% your own participation, measured from what you posted and how you took part in your team's ceremonies.

Your team's twice-weekly stand-ups are graded inside the sprint, not as a separate assignment. They are scheduled on your own team's availability once you are placed. They are not attendance in the registrar's sense — they are how the sprint is worked.

The participation share is yours alone. A strong artifact cannot carry a member who did not take part, and a difficult team cannot sink a member who did.

What you are assessed against

This course is where Student Outcomes 1, 2 and 6 are assessed on your own work, and where Outcomes 3, 4 and 5 are evaluated at the showcase and across your team's term. You are assessed against your own degree program's wording of these outcomes — Computer Science, Cybersecurity, Information Technology, or Data Science & AI — including when your team is building a project that was originally proposed for a different degree. Projects are recast to fit the team's own discipline in week two, and that is what the team is then measured on.

Each sprint produces a named part of that evidence, in your degree's own terms, and your sprint page tells you which part each time. By the end of Sprint 5 you will have produced every outcome item your program is measured against — once each — without ever having been handed a checklist.

The two December events

Friday 4 December — the Capstone Showcase. Every team presents. Industry judges score the work live on a seven-item form, and all four awards are announced at the closing ceremony that afternoon. It is in person. Students in the in-person section are required to attend in person; students in the online section are strongly encouraged to attend but are not required to be there.

Friday 11 December — the Capstone 1 Virtual Showcase. One week after you present, you are the judge. Each of you evaluates three Capstone 1 projects on the same five presentation items you were judged on — Domain Knowledge, Organization, Presentation Aids, Elocution, Audience Contact — each rated 1 to 5, with one sentence of comment per project. It is worth 30 points, 10 per evaluation, and it is online, so nobody travels for it. Your three evaluations are due Sunday 13 December.

Neither showcase is the deadline. The awards are settled on 4 December and nothing carries over. **Your final deliverable is due Monday 14 December,** it is submitted once per team by the team leader, and it decides no award — it is the archive.

Deadlines and the weekend grace

The course week runs Monday to Friday, and work is due Friday at 11:59 PM Eastern.

If your week went wrong, take the weekend. Anything posted by Sunday at 11:59 PM Eastern counts in full, with no penalty. **You do not have to ask and you do not have to explain.** Work is graded on Monday mornings.

Four dates have no weekend, because they are events or fixed publications rather than deadlines: Showcase Preparation (Wed 2 Dec — Capstone 1 students evaluate from your materials and need them published), the Capstone Showcase (Fri 4 Dec), the Capstone 1 Virtual Showcase (Fri 11

Dec), and the Final Deliverable (Mon 14 Dec).

If you cannot finish even with the weekend, something real has happened and I want to know. **Send me a message in Canvas Inbox.** Please do not email me about this course.

Academic integrity in your submitted work

You are expected to use AI tools in this course. Using them is not cheating. Concealing that you used them is. Every submission carries a four-line record: which tools you used and for what, what you kept, what you checked, and what the tool got wrong. A record that never finds an error is evidence you are not checking.

In week 13 your team files a **Provenance & Ethics Clearance** covering the whole project — what came from where, what licence it carries, what data was used, and what was verified. It is worth 40 points and it ships inside your final package.

Grade scale

A 930–1000 · **A–** 900–929 · **B+** 870–899 · **B** 830–869 · **B–** 800–829 · **C+** 770–799 · **C** 700–769 · **D** 600–699 · **F** below 600

Grading

Grading

#	Assignment	Pts	Published due	Grace to (Sun)	Scored from
1	W1 · Course Orientation Wizard	30	Fri 28 Aug	Sun 30 Aug	15 pages x 2 each

#	Assignment	Pts	Published due	Grace to (Sun)	Scored from
2	W1 · Intake Wizard	20	Fri 28 Aug	Sun 30 Aug	10 sections x 2 each
3	W1 · Project Review & Bid Wizard	60	Fri 28 Aug	Sun 30 Aug	Per project reviewed
4	W1 · Team Formation Wizard	40	Fri 28 Aug	Sun 30 Aug	Path complete
5	W2 · Cast the Problem & Semester Plan <i>(team)</i>	60	Fri 4 Sep	Sun 6 Sep	60 / 40
6	W3-4 · Sprint 1 — <i>Is the problem real, and can we run?</i>	50	Fri 18 Sep	Sun 20 Sep	60 / 40
7	W5-6 · Sprint 2 — <i>What are we building, and why this way?</i>	75	Fri 2 Oct	Sun 4 Oct	60 / 40

#	Assignment	Pts	Published due	Grace to (Sun)	Scored from
8	W7-8 · Sprint 3 — <i>Does the core thing work?</i>	100	Fri 16 Oct	Sun 18 Oct	60 / 40
9	W9-10 · Sprint 4 — <i>Is it any good, measured against a criterion?</i>	125	Fri 30 Oct	Sun 1 Nov	60 / 40
10	W11-12 · Sprint 5 — <i>Can someone else run it, and are we honest?</i>	150	Fri 13 Nov	Sun 15 Nov	60 / 40
11	W13 · Provenance & Ethics Clearance	40	Fri 20 Nov	Sun 22 Nov	Cleared list + verified refs

#	Assignment	Pts	Published due	Grace to (Sun)	Scored from
12	W14 · Showcase Preparation	20	Wed 2 Dec	—	Poster · slides · video · rehearsal, 5 each
13	W15 · Capstone Showcase — present	100	Fri 4 Dec (event)	—	Judges' scores
14	W16 · Capstone 1 Virtual Showcase — judge three projects	30	Fri 11 Dec (event)	Sun 13 Dec (evaluations)	3 evaluations x 10 each
15	Final Deliverables — team leader submits	100	Mon 14 Dec	—	One team submission
	Total	1000			

Sprints are scored in two parts: the team's sprint artifact, and your own participation measured from what you posted and how you took part in the ceremonies. The participation share is yours alone — a strong artifact cannot carry a member who did not take part, and a difficult team cannot sink a member who did.

The five sprints ramp: 50 · 75 · 100 · 125 · 150. Sprint 5 is worth three times Sprint 1. All five count and none is dropped — the ramp is the forgiveness, and it is front-loaded on purpose, because Sprint 1 is the sprint you are least equipped to do well.

Grade scale: A 930-1000 · A- 900-929 · B+ 870-899 · B 830-869 · B- 800-829 · C+ 770-799 · C 700-769 · D 600-699 · F below 600.

Late work

The course week runs Monday to Friday, and everything is due Friday at 11:59 PM Eastern.

If your week went wrong, take the weekend. Anything posted by Sunday at 11:59 PM Eastern counts in full, with no penalty. You do not have to ask and you do not have to explain.

Work is graded on Monday mornings. Four dates have no weekend because they are events or fixed publications rather than deadlines: Showcase Preparation, Wednesday 2 December (Capstone 1 students evaluate from your materials and need them published), the Capstone Showcase, Friday 4 December, the Capstone 1 Virtual Showcase, Friday 11 December, and the Final Deliverable, Monday 14 December. Your three evaluations from 11 December are themselves due Sunday 13 December, with the normal grace.

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[Urgent mobile contact removed from public copy]

Proctored Exams

No proctored exam.

Canvas Schedule

Due Date	Assignment Name	Assignment Type	Points
	Academic Honesty Policy	Quiz	0
	Practice Quiz	Quiz	0
8/28/26	W1 · Course Orientation Wizard	Assignment	30
8/28/26	W1 · Intake Wizard	Assignment	20
8/28/26	W1 · Project Review & Bid Wizard	Assignment	60
8/28/26	W1 · Team Formation Wizard	Assignment	40
9/4/26	W2 · Cast the Problem & Semester Plan	Assignment	60
9/18/26	Sprint 1 · Is the problem real, and can we run?	Assignment	50
10/2/26	Sprint 2 · What are we building, and why this way?	Assignment	75
10/16/26	Sprint 3 · Does the core thing work?	Assignment	100
10/30/26	Sprint 4 · Is it any good, measured against a criterion?	Assignment	125
11/13/26	Sprint 5 · Can someone else run it, and are we honest?	Assignment	150

Due Date	Assignment Name	Assignment Type	Points
11/20/26	W13 · Provenance & Ethics Clearance	Assignment	40
12/1/26	W14 · Showcase Preparation	Assignment	20
12/3/26	W15 · Capstone Showcase — present	Assignment	100
12/13/26	W16 · Capstone 1 Virtual Showcase — judge three projects	Assignment	30
12/14/26	Final Deliverables — team leader submits	Assignment	100

Schedule of Topics

A note on what this schedule is. There are no class meetings in this course — no lectures, no synchronous sessions, in person or online, in any week of the term. The "topic" column below is what your team is doing that week, not a session you attend. The only things on your calendar are your team's own twice-weekly stand-ups, which your team schedules itself, and the Capstone Showcase on Friday 4 December.

Everything is due Friday at 11:59 PM Eastern, with a silent grace to Sunday at 11:59 PM Eastern — except the four dates marked "no grace," which are events or fixed publications.

Week	Dates	What your team is doing	Due that week
1	24–28 Aug	Week one. Orientation, intake, reviewing every project on offer, and telling us how you want to be placed.	All four wizards — Fri 28 Aug — 150 pts
2	31 Aug – 4 Sep	Teams and projects announced Monday 31 August. Meet your Product Owner. Cast the problem into your own discipline's terms and plan the five sprints.	Cast the Problem & Semester Plan — Fri 4 Sep — 60 pts
3	7–11 Sep	Sprint 1 begins. Establish the cadence. Test the cast problem against your Product Owner and against the real data or environment.	—

Week	Dates	What your team is doing	Due that week
4	14–18 Sep	Sprint 1 closes. A walking skeleton that runs end to end — it proves the path exists, not that it is good yet. Sprint review and retrospective.	Sprint 1 — Fri 18 Sep — 50 pts
5	21–25 Sep	Sprint 2 begins. Choose the approach. Record the alternatives you rejected and the criterion you used to choose.	—
6	28 Sep – 2 Oct	Sprint 2 closes. Requirements and design written down. Sprint review and retrospective.	Sprint 2 — Fri 2 Oct — 75 pts
7	5–9 Oct	Sprint 3 begins. Build the central capability. The most build-heavy stretch of the term.	—

Week	Dates	What your team is doing	Due that week
8	12–16 Oct	Sprint 3 closes. The core thing runs end to end on real public or synthetic inputs. Sprint review and retrospective.	Sprint 3 — Fri 16 Oct — 100 pts
9	19–23 Oct	Sprint 4 begins. Set up the measurement — against the criterion you stated in advance, not one chosen afterwards.	—
10	26–30 Oct	Sprint 4 closes. Evaluate. Report what the measurement says, including where it disagrees with what you hoped.	Sprint 4 — Fri 30 Oct — 125 pts

Week	Dates	What your team is doing	Due that week
11	2–6 Nov	Sprint 5 begins. Harden the system. Make it something another person could pick up and operate.	—
12	9–13 Nov	Sprint 5 closes. Hand-over, continuity, and the honest limitations statement. The largest sprint of the term.	Sprint 5 — Fri 13 Nov — 150 pts
13	16–20 Nov	Provenance and ethics. Every third-party component, dataset, model and asset, with its source and licence. Every reference verified.	Provenance & Ethics Clearance — Fri 20 Nov — 40 pts

Week	Dates	What your team is doing	Due that week
14	23–27 Nov	Thanksgiving week. Nothing is due. Use it to build your poster, slides and demo video if you want to be comfortable in week 15.	—
15	30 Nov – 4 Dec	Showcase week. Publish your materials Wednesday; present Friday.	Showcase Preparation — Wed 2 Dec — 20 pts — no grace Capstone Showcase — Fri 4 Dec — 100 pts — no grace, in person
16	7–11 Dec	You are the judges. Evaluate three Capstone 1 projects at their virtual showcase on Friday. Online — nobody travels for it.	Capstone 1 Virtual Showcase — Fri 11 Dec — 30 pts — event, no grace <i>(your three evaluations are due Sun 13 Dec)</i>

Week	Dates	What your team is doing	Due that week
—	Mon 14 Dec	Final deliverable. One submission per team, made by the team leader.	Final Deliverables — Mon 14 Dec — 100 pts — no grace
—	Wed 16 Dec	Final grades posted.	—

The five dates that do not move

- **Monday 31 August** — teams and projects announced
- **Friday 4 December** — Capstone Showcase, in person, every team presents
- **Friday 11 December** — Capstone 1 Virtual Showcase, where you are the judges
- **Sunday 13 December** — your three evaluations due
- **Monday 14 December** — final deliverable

Neither December showcase is the deadline. All four awards are announced at the closing ceremony on 4 December and nothing carries over. The archive you submit on 14 December decides no award.

Two weeks worth planning around

Week one is 150 points and it cannot be made up. Placement runs on Monday 31 August from what you submitted the week before. It is the most expensive week in the course to treat as a slow start.

Week 14 is Thanksgiving and nothing is due. That is deliberate — it is the only breathing room in the second half of the term, and the teams that use it to get their showcase materials done are the teams that enjoy week 15.

Official Catalog Course Description

Students work on faculty supervised projects in teams of up to 5 members to design and implement solutions to problems utilizing knowledge obtained across the spectrum of Computer Science courses.

Official Course Outcomes

Outcome	Degree Outcome(s)	ABET Topic(s)
Demonstrate mastery of problem formulation.	B* 1	TECH
Demonstrate mastery of specifying the requirements of a problem.	B* 1	CBS, SWEG
Demonstrate mastery of designing the solution to a problem.	B* 2	CBS, SWEG
Demonstrate mastery of realizing the solution to a problem.	B* 2	CBS, SWEG
Demonstrate the ability to validate and evaluate the solution to a problem.	B* 2	CBS, SWEG
Demonstrate the ability to manage a semester long project.	BS-* 6	INFO, PROJ
Demonstrate the ability to work effectively in a project team.	B* 5	PROJ
Demonstrate the ability to think logically and critically when developing the solution to a given problem.	BS-* 6	TECH
Demonstrate the ability to apply concepts learned in various courses when developing the solution to a given problem.	BS-* 6	TECH
Demonstrate the ability to communicate the details of the technical solution through verbal and written modes.	B* 3	TECH

Demonstrate the ability to incorporate ethical issues into the project development and documentation process.	B* 4	IMP
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